Dataset Documentation

FinSight AI – Customer Financial Analytics & Risk Prediction

1. Dataset Overview

Dataset Name: synthetic_personal_finance_dataset
Source: Kaggle
Dataset URL: https://www.kaggle.com/datasets/miadul/personal-finance-ml-dataset
Format: CSV
Type: Synthetic Personal Finance Dataset

2. Dataset Statistics

Raw Dataset Rows: 32,424
Raw Dataset Columns: 20
Final Dataset Rows: 32,424
Final Dataset Columns: 22 (including Risk_Score and Risk_Category)

3. Dataset Features

user_id
age
gender
education_level
employment_status
job_title
monthly_income_usd
monthly_expenses_usd
savings_usd
has_loan
loan_type
loan_amount_usd
loan_term_months
monthly_emi_usd
loan_interest_rate_pct
debt_to_income_ratio
credit_score
savings_to_income_ratio
region
record_date
Risk_Score
Risk_Category

4. Data Cleaning

• Checked dataset structure and data types.
• Verified duplicate records.
• Filled missing values in **loan_type** with **'No Loan'**.
• Confirmed final missing values = 0.
• Prepared cleaned dataset for machine learning.

5. Feature Engineering

Two new columns were created:
• Risk_Score – Numerical financial risk score.

• Risk_Category – Low, Medium and High financial risk classes.

6. Machine Learning Preparation

Preprocessing included:
• OneHotEncoder for categorical variables.
• StandardScaler for numerical variables.
• Train-Test Split with stratification.
• Pipeline and ColumnTransformer implementation.

7. Model Performance

Decision Tree – Accuracy: 96.56%, Precision: 96.56%, Recall: 96.56%, F1 Score: 96.56%
Random Forest – Accuracy: 94.39%, Precision: 94.55%, Recall: 94.39%, F1 Score: 94.41%
KNN – Accuracy: 85.38%, Precision: 85.82%, Recall: 85.38%, F1 Score: 85.48%

8. Conclusion

The cleaned dataset was successfully prepared for machine learning. Missing values were handled, new features (Risk_Score and Risk_Category) were created, and the dataset was used to train Decision Tree, Random Forest and KNN models for customer financial risk prediction.